THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS V — MATHEMATICS · MODEL PAPER 06 ANSWER KEY

Answers and marking points | Maximum Marks: 80

UNDERSTANDING SHAPE AND SPACE — Marks 21

Q.	Marks	Answer / marking points
1	1	Each small square of the figure is 1 square cm. What is the area of the shaded figure? **(c) 17 sq cm**
2	6	**(i)** Look at the three angles drawn. Write the type of each angle… [3] (a) acute (less than 90°) (b) straight angle (c) reflex (more than 180°) **(ii)** (a) The minute hand of a clock moves from 12 to 3. Through h… [3] (a) 90° (a right angle) (b) 180°, a straight angle (c) 360 ÷ 90 = 4
3	6	**(i)** Write the number asked: (a) edges of a cuboid (b) faces of a… [3] (a) 12 (b) 5 (c) 5 **(ii)** Name the solid that each net folds into: (a) a net made of 6… [3] (a) cube (b) cone (c) cylinder
4	8	The plan of a house in Sikar is given below (not to scale; the north direction is shown). **(a)** What is the area of the Hall? [2] 6 × 5 = 30 sq m **(b)** Which room has the smallest area? Write its area. [2] Store: 4 × 3 = 12 sq m (Kitchen 20, Bedroom 18, Hall 30) **(c)** Find the perimeter of the Bedroom. [2] 2 × (6 + 3) = 18 m **(d)** The Verandah is to be tiled at ₹ 150 per sq m. Find the cost… [2] Area = 10 × 2 = 20 sq m; cost = 20 × 150 = ₹ 3,000

NUMBER SYSTEM — Marks 9

Q.	Marks	Answer / marking points
5	1	In the number 4,07,285 the value of the digit 7 is **(b) 7000**
6	1	The greatest 6-digit number that can be made from the digits 3, 0, 8, 5, 1, 9 (each used o… **(a) 9,85,310**
7	1	300000 + 40000 + 500 + 6 is the expanded form of **(d) 3,40,506**
8	6	**(i)** A water tank holds 1000 litres. On Monday 3/10 of it was fil… [3] (a) 3/10 + 2/5 = 3/10 + 4/10 = **7/10** (b) 1 − 7/10 = **3/10** (c) 3/10 of 1000 = **300 litres** **(ii)** The number line from 0 to 1 is divided into 8 equal parts. W… [3] A = 1/8 B = 4/8 = 1/2 C = 7/8

OPERATIONS — Marks 25

Q.	Marks	Answer / marking points
9	1	The product of 405 and 12 is **(c) 4,860**
10	3	A dairy in Jaipur filled 24 cans with 15 litres of milk in each can. In the morning 280 li… Total milk = 24 × 15 = 360 L Remaining = 360 − 280 = 80 L Packets = 80 ÷ 2 = **40**
11	3	Divide 5,843 by 7. Write the quotient and the remainder and check your answer. Quotient = **834**, remainder = **5**. Check: divisor × quotient + remainder = 7 × 834 + 5 = 5838 + 5 = 5843, which is the dividend. ✓
12	6	**(i)** The Sharma family spends in one week: vegetables ₹ 450, milk… [3] (a) 450 + 280 + 620 + 350 = ₹ 1,700 (b) 1,700 × 4 = ₹ 6,800 (c) 9,500 − 6,800 = ₹ 2,700 **(ii)** Use a short method: (a) 73 × 99 (b) 46 × 25 (c) 8 × 125 [3] (a) 73 × 100 − 73 = 7300 − 73 = **7227** (b) 46 × 100 ÷ 4 = 4600 ÷ 4 = **1150** (c) 8 × 125: 8 × 100 = 800 and 8 × 25 = 200, so 800 + 200 = **1000**

Q.	Marks	Answer / marking points
13	6	Bulbs numbered 1 to 24 are fixed on a board. Switch A lights every 3rd bulb (3, 6, 9, …) a… **(i)** (a) Which bulbs are lit by both the switches? (b) How many b… [3] (a) **12 and 24** (green) (b) 4, 8, 12, 16, 20, 24 = **6** bulbs (c) LCM of 3 and 4 = 12 **(ii)** (a) Find the LCM of 6 and 8. (b) Find the HCF of 12 and 18. … [3] (a) **24** (b) factors of 12: 1,2,3,4,6,12; of 18: 1,2,3,6,9,18; HCF = **6** (c) together every 24 s; 120 ÷ 24 = **5** times
14	6	**(i)** The square on the right is a magic square. (a) What is the s… [3] (a) 12 + 5 + 10 = **27** (b) 12 + 9 + 6 = **27** (c) 27 ÷ 3 = **9** (the middle number) **(ii)** 48 square tiles, each of side 1 m, are arranged to make a re… [3] Factor pairs of 48: 1×48, 2×24, 3×16, 4×12, 6×8. (a) **5** ways (b) **8 m by 6 m** (c) 2 × (8 + 6) = **28 m**

MEASUREMENT — Marks 17

Q.	Marks	Answer / marking points
15	3	Fill in the blanks: (a) 5 km 40 m = _____ m (b) 3 kg 250 g = _____ g (c) 3 hours 15 … (a) 5040 (b) 3250 (c) 195
16	3	Write True or False: (a) 6:30 pm is written as 18:30 in the 24-hour clock. (b) 450 g is mo… (a) True (6 + 12 = 18) (b) False (half kg = 500 g) (c) False (about 37°C)
17	3	A train leaves Jaipur at 9:45 am and reaches Kota at 2:20 pm the same day. How long does t… 2:20 pm = 14:20. From 9:45 to 14:20: 9:45 → 12:00 = 2 h 15 min; 12:00 → 14:20 = 2 h 20 min. Total = **4 hours 35 minutes**
18	8	The road map of some cities of Rajasthan is shown (distances are given in km). **(a)** A bus goes from Alwar to Ajmer through Jaipur. How far does … [2] 150 + 130 = **280 km** **(b)** Which route from Jaipur to Chittorgarh is shorter — through … [2] Through Ajmer: 320 km; through Kota: 420 km. Through **Ajmer** is shorter by 100 km. **(c)** A car runs 5 km on 1 litre of petrol. How many litres are ne… [2] Distance = 150 + 250 = 400 km; petrol = 400 ÷ 5 = **80 litres** **(d)** Which city is nearest to Jaipur? Write the distance. [2] **Ajmer**, 130 km (Alwar 150, Kota 250)

DATA HANDLING AND PATTERNS — Marks 8

Q.	Marks	Answer / marking points
19	1	Which of the following is a square number? **(b) 144**
20	1	Squares are arranged in steps as shown. How many squares will be there in Fig. 6? **(a) 21 — 1, 3, 6, 10, 15, 21**
21	3	The bar graph shows the favourite game of the students of Class 5. **(a)** Which game is liked by the most students? [1] Cricket (24) **(b)** How many more students like Cricket than Hockey? [1] 24 − 6 = 18 **(c)** How many students are there in all? [1] 24 + 18 + 12 + 6 + 15 = 75
22	3	Look at the pattern: 5, 9, 13, 17, ___, ___. (a) Write the rule. (b) Write the next two nu… (a) add 4 each time (b) 21, 25 (c) 5 + 9 × 4 = 41

END OF ANSWER KEY — Total: 80 Marks